

Animal Farm Management System (AFMS) Software Requirements Specification (SRS)

1. Project Overview

The Animal Farm Management System is a cross-platform mobile application developed using React Native for Android and iOS. It helps farm managers and employees manage daily operations including animal care, maintenance, task tracking, and issue reporting.

2. User Roles

A. Manager - Login using email and password. - Create employee accounts and provide credentials. - Assign work and checklists. - Monitor task completion status. - View uploaded photos as proof of work. - Receive instant notifications for reported issues. - View analytics and export PDF reports.

B. Employee - Login using username and password. - View assigned farm tasks. - Complete checklist items. - Upload task completion photos. - Report problems using title, description, photos, and audio recordings.

3. Farm Categories - Birds - Fish Tanks - Cows - Calves - Vehicles

4. Functional Modules

Authentication: Firebase Authentication with role-based access.

Task Management: Managers assign tasks. Employees update progress and submit proof images. Updates are synchronized in real time.

Issue Reporting: Employees can submit issues with: - Issue title - Description - Photos - Audio notes

Managers receive push notifications when a new issue is created.

5. Technology Stack

Frontend: - React Native - Expo - TypeScript - React Navigation - NativeWind or React Native Paper

Backend: - Firebase Firestore Database - Firebase Storage - Firebase Cloud Messaging - Node.js/Express for advanced APIs if required

6. Database Collections

Users: Manager and employee details with roles.

Tasks: Category, assigned employee, status, images, timestamps.

Issues: Title, description, media attachments, reporter details, status, and date.

7. User Interface Design

Theme: Modern nature-inspired design using: - Forest Green - Earth Brown - Sky Blue - Soft Cream

Design principles: - Clean dashboard - Rounded cards - Smooth animations - Large touch-friendly buttons

8. Development & Testing

IDE: Visual Studio Code.

Testing Methods: - Expo Go during early development. - Android Emulator via Android Studio. - Physical Android device using USB debugging. - iOS simulator using Xcode or cloud builds.

9. Future Enhancements

- GPS-based task verification - Offline functionality - AI animal disease detection - Voice commands
- Multi-language support - Inventory and medicine management

Conclusion: The AFMS is designed as a scalable, professional farm operations platform that improves communication between managers and employees while maintaining real-time records and reports.